The Effect of Background Music on Focused Tasks: A Nested and Blocked Experimental Design

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Abstract

This study examines the impact of the features of background music (BGM) on cognitive functioning during focused tasks of different type (verbal vs. mathematical) and level of difficulty (easy vs. hard). Background music type (instrumental vs. with lyrics) and volume of music playback (0%, 50%, 100%) were manipulated. We completed a total of 288 trial sessions in a nested and blocked design with volume nested in music type and with participants and time blocks treated as blocking variables. Task completion time was the main outcome. Linear mixed-effects models indicated all main effects of music type, volume, task type, and difficulty on performance time, including a significant task type by difficulty interaction. The instrumental music at a volume of 50% resulted in the shortest finishing time and significantly better performance than all others including silence. Diagnostics confirmed model assumptions were met to a satisfactory level. These results indicate a potential cognitive efficiency and performance boost by using instrumental music at a volume of 50%, providing practical recommendations on how to optimize audio environments in the workplace.

Keywords: Nested Design, Background Music, Cognitive Tasks

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1 Introduction

1.1 Motivation

In modern work and study environments, background music (BGM) is commonly used with the intention of improving focus, mood, or productivity. However, whether BGM genuinely enhances cognitive performance — or becomes a source of distraction — remains an open question. Previous research suggests that the impact of BGM may depend on several factors, including whether music is present at all, whether it contains lyrics, and how loud it is. In addition, the type of task being performed (verbal or computational) can interact with these audio conditions in complex ways. Our experiment seeks to explore these relationships by testing how the presence or absence of BGM, along with its specific characteristics (music type and volume), influences task completion time for both verbal and arithmetic tasks. By including a no-music condition alongside lyrical and instrumental BGM, this study aims to clarify whether silence is truly golden, or whether certain types of music can reliably support cognitive efficiency. The findings have practical implications for individuals trying to optimize their focus in academic, professional, or personal settings.

1.2 Research Questions

We have two main research questions to figure out.

- What factors significantly influence focused task completion time?
- What combination of background music conditions minimizes task completion time, thereby maximizing cognitive efficiency?

2 Experimental Design

2.1 Factors and Blocks

We have following factors in our experiment:

- **Music Type**: A fixed effect with two levels:
  - Inst: Instrumental music (no lyrics)
  - Lyrics: Music with lyrics
- **Volume**: A fixed effect nested within Music Type, with three levels:
  - 0%: Silence
  - 50%: Half volume level
  - 100%: Full volume level

Music Type nests Volume because the way listeners perceive and respond to volume varies between instrumental and lyrical music. This arrangement accepts that volume effects must be evaluated within their respective music types because direct comparison between them is not feasible.

- **Task Type**: A fixed effect with two levels:
  - Verbal: Filling in one/two blanks to make a word
  - Math: Two-digit mix or multiply
- **Task Difficulty**: A fixed effect with two levels:
  - Easy: Simpler tasks (4-letter words or two-digit mix)
  - Hard: More challenging tasks (6-7 letter words or two-digit multiplication)
- **Time Interval**: A random block effect with three levels:
  - T1: April 22, 2025, 21:30–23:00
  - T2: April 24, 2025, 12:30–14:00
  - T3: April 27, 2025, 17:00–18:30

  These specific time intervals were randomly selected from the broader experimental period (April 21 to April 27, 2025, from 8:00 AM to 11:59 PM). The complete experimental period was segmented into sequential 90-minute time blocks and three blocks were chosen at random to represent various daytime periods. We modeled Time Interval as a random effect because the selected three periods represent a sample of all possible time periods which lets us extend our results across various times of day while minimizing circadian fluctuations in attention and fatigue.
- **Participant**: A fixed block effect with two levels ($n = 1, 2$)
  - P1: Xin Liu
  - P2: Haiyang Qin

  We implemented a standardized experimental protocol to ensure consistency across all trials and minimize potential confounding variables.

2.2 Protocols

2.2.1 Task Generation

All tasks were generated in advance using standardized prompts to ensure a uniform difficulty level across trials.

Prompts:

- **Verbal – Easy:**
  Select a 5–8 letter TOEFL-level English word. Replace *exactly one* letter with "_".

- **Verbal – Hard:**
  Select a 6–10 letter TOEFL-level English word. Replace *exactly two letters* with "_".

- **Math – Easy:**
  Generate 5 subtraction or addition questions and 5 multiplication questions using two random two-digit integers such that the result is non-negative.

- **Math – Hard:**
  Generate a problem involving multiplication of two random two-digit positive integers.

Rules

- A trial contains exactly ten questions.

- Every word and arithmetic expression must be unique within the booklet.

Verbal Task Examples:

Table 1: Verbal Tasks - Easy Level

No.	Verbal-Easy
1	s_bling
2	exte_nal
3	sei_e
4	am_ze
5	p_rceive
6	wa_er
7	ho_se
8	_lock
9	ro_m
10	br_ad

Table 2: Verbal Tasks - Hard Level

No.	Verbal-Hard
1	th_ra_y
2	ca_e_r
3	_ecorat_
4	numer_u_
5	r_ndo_
6	comp_ex_ty
7	algo_ith_
8	_nalyt_cs
9	inte_act_on
10	stat_stic_

Math Task Examples:

Table 3: Math Tasks - Easy Level

No.	Math-Easy
1	67 - 50
2	33 + 52
3	87 + 67
4	97 + 73
5	38 - 33
6	45 * 28
7	76 * 29
8	64 * 18
9	93 * 46
10	55 * 37

Table 4: Math Tasks - Hard Level

No.	Math-Hard
1	58 * 37
2	37 * 11
3	97 * 70
4	38 * 35
5	27 * 32
6	65 * 28
7	84 * 67
8	73 * 28
9	92 * 47
10	62 * 15

The tasks were prepared and assigned to trials based on the required task type and difficulty level. Each treatment combination was repeated twice to increase the reliability of our findings.

2.2.2 Music Selection and Volume Control

For the background music conditions, we selected:

- **Instrumental (Inst):** A Broken Heart Heals with Time
- **Lyrics:** Loving Strangers

The volume levels were quantified and calibrated consistently across all trials:

- 0%: Silence
- 50%: Half of the MacBook Volume
- 100%: Full of the MacBook Volume

2.2.3 Experimental Procedure

For each trial:

1. The participant was presented with the appropriate task based on the randomized schedule.
2. Background music was played at the designated volume level.
3. The participant completed the task while timing was recorded.
4. Task completion time was measured in seconds, recorded to two decimal places (e.g., 48.55 seconds).

2.3 Randomization

To prevent systematic bias and reduce the impact of lurking order effects, we randomized the run order of all treatment trials. The full factorial design included seventy-two unique treatment combinations ($2 * 3 * 2 * 2 = 24$), and each was replicated twice for each of the two experimenters in each time block ($24 * 2 * 3 * 2 = 288$), yielding a total of 288 trials. The randomization was performed in R using a fixed random seed (set.seed(42)) to ensure reproducibility. The resulting randomized sequence was used to guide the actual order of experimentation, independent of the structure of the design matrix.

To account for potential variability in participants' performance due to the time of day, we included Time Interval as a random block in our experimental design. The overall experimental period spanned from April 21 to April 27, 2025, covering daily hours from 8:00 AM to 11:59 PM. This range was divided into consecutive 90-minute blocks. From this full set of possible time blocks, we randomly selected three intervals to represent different times of day: T1 (April 22, 21:30–23:00), T2 (April 24, 12:30–14:00), and T3 (April 27, 17:00–18:30). By treating these as a sample from a broader population of possible time periods, we modeled Time Interval as a random block effect to better generalize our findings across various times of day and reduce confounding due to circadian fluctuations in attention or fatigue.

3 Statistical Analysis

3.1 Model Specification

We model the task completion time Y_{ijklmn} using the following mixed-effects model. The initial model is the following:

$$\begin{aligned} Y_{ijklmno} = \mu + \alpha_i + \beta_{j(i)} + \gamma_k + \delta_l + \phi_n + \tau_m \\ + (\alpha\gamma)_{ik} + (\alpha\delta)_{il} + (\gamma\delta)_{kl} + (\alpha\gamma\delta)_{ikl} + \varepsilon_{ijklmno} \end{aligned}$$

Where:

- $Y_{ijklmno}$: task completion time under block n and factor i, j, k, l, m
- μ: overall mean
- α_i: fixed effect of Music Type ($i = 1, 2$)
- $\beta_{j(i)}$: fixed effect of Volume level ($j = 1, 2, 3$), nested within Music Type
- γ_k: fixed effect of Task Type ($k = 1, 2$)
- δ_l: fixed effect of Task Difficulty ($l = 1, 2$)
- ϕ_n: fixed block effect of Participant ($n = 1, 2, 3$)
- $\tau_m \sim \mathcal{N}(0, \sigma_\tau^2)$: random block effect of Time Interval ($m = 1, 2, 3$)
- $\varepsilon_{ijklmno}$: random error term

We include selected two-way and three-way interaction terms among the fixed factors Music Type (α), Task Type (γ), and Difficulty (δ) to investigate how background music may interact with cognitive task demands. Such interactions are necessary to explore whether the impact of music is contingent upon task type or difficulty. In particular, the $\alpha\gamma\delta$ three-way interaction permits us to examine whether effects of music type change simultaneously with task type and difficulty level.

We do not include any interactions involving Volume (β), which is nested within Music Type, because nested factors are not comparable across levels of their parent factor. Interactions involving β would introduce the nested structure and lead to interpretational ambiguity. Similarly, we exclude interactions involving Participant (ϕ) and Time Block (τ) since these are block-related effects intended to control for subject-level and temporal variability, not to test treatment interaction hypotheses. Including block interactions would conflate experimental control with treatment effects, which is not aligned with our research goals.

The following model assumptions and identifiability constraints are imposed:

- Independence: Time τ_m is independent of $\varepsilon_{ijklmno}$

- Homoscedasticity: residuals have constant variance
- Normality: $\varepsilon_{ijklmno} \sim \mathcal{N}(0, \sigma^2)$
- Sum-to-zero constraints:

$$\sum_i \alpha_i = 0 \quad \sum_j \beta_{j(i)} = 0 \quad \text{(for each } i\text{)}$$
$$\sum_k \gamma_k = 0 \quad \sum_l \delta_l = 0 \quad \sum_n \phi_n = 0$$

All interaction terms sum to 0 over any index when others are fixed.

3.2 Model Selection

We start with testing the significance of the three-way interaction.

$$\begin{cases} H_0 : (\alpha\gamma\delta)_{ikl} = 0 \quad \text{for all } i, k, l \\ H_A : \text{Not all } (\alpha\gamma\delta)_{ikl} = 0 \end{cases}$$

Based on Figure 1, the F-statistic for the three-way interaction $\alpha\gamma\delta$ is 1.38 with a p-value of 0.2418. Therefore, we fail to reject the null hypothesis at any reasonable significance level. Thus, there is no significant three-way interaction among Music Type, Task Type, and Difficulty.

The GLM Procedure
Tests of Hypotheses for Mixed Model Analysis of Variance

Dependent Variable: Time Time

	Source	DF	Type III SS	Mean Square	F Value	Pr > F
*	MusicType	1	2696.472006	2696.472006	722.40	<.0001
	Volume(MusicType)	4	1827.405819	456.851455	122.39	<.0001
*	TaskType	1	1243.259113	1243.259113	333.08	<.0001
*	Difficulty	1	3088.980000	3088.980000	827.56	<.0001
	Participant	1	1.193512	1.193512	0.32	0.5722
	TimeInterval	2	140.248252	70.124126	18.79	<.0001
*	MusicType*TaskType	1	0.568889	0.568889	0.15	0.6965
*	MusicType*Difficulty	1	0.248513	0.248513	0.07	0.7966
*	TaskType*Difficulty	1	185.730689	185.730689	49.76	<.0001
	MusicT*TaskTy*Diffic	1	5.136012	5.136012	1.38	0.2418
	Error: MS(Error)	273	1019.011683	3.732644		

* This test assumes one or more other fixed effects are zero.

Figure 1: Full Model ANOVA Table

Then we remove the highest order interaction and fit a new model that only include two-way interactions. We now test the significance of the remaining two-way interaction terms.

$$\begin{cases} H_0 : (\alpha\gamma)_{ik} = 0 \quad \text{for all } i, k \\ H_A : \text{Not all } (\alpha\gamma)_{ik} = 0 \end{cases} \qquad \begin{cases} H_0 : (\alpha\delta)_{il} = 0 \quad \text{for all } i, l \\ H_A : \text{Not all } (\alpha\delta)_{il} = 0 \end{cases} \qquad \begin{cases} H_0 : (\gamma\delta)_{kl} = 0 \quad \text{for all } k, l \\ H_A : \text{Not all } (\gamma\delta)_{kl} = 0 \end{cases}$$

	Source	DF	Type III SS	Mean Square	F Value	Pr > F
*	MusicType	1	2696.472006	2696.472006	721.41	<.0001
	Volume(MusicType)	4	1827.405819	456.851455	122.23	<.0001
*	TaskType	1	1243.259113	1243.259113	332.62	<.0001
*	Difficulty	1	3088.980000	3088.980000	826.42	<.0001
	Participant	1	1.193512	1.193512	0.32	0.5725
	TimeInterval	2	140.248252	70.124126	18.76	<.0001
	MusicType*TaskType	1	0.568889	0.568889	0.15	0.6967
	MusicType*Difficulty	1	0.248513	0.248513	0.07	0.7967
	TaskType*Difficulty	1	185.730689	185.730689	49.69	<.0001
	Error: MS(Error)	274	1024.147695	3.737765		
* This test assumes one or more other fixed effects are zero.						

Figure 2: Two-way Interaction Model ANOVA Table

The test for the interaction between Music Type and Task Type yields a p-value of 0.6967. Since the p-value is greater than 0.05, we fail to reject H_0. The test for the interaction between Music Type and Difficulty gives a p-value of 0.7967. Again, we fail to reject H_0. However, the interaction between Task Type and Difficulty is statistically significant, with an F-value of 49.69 and a p-value < 0.0001. Therefore, we reject H_0 for this interaction.

The significant interaction between Task Type and Difficulty suggests that the effect of task difficulty on performance time differs depending on the type of task (verbal or math). Practically, this means that increasing difficulty does not uniformly increase task time across both task types. This is due to the character of the tasks: whereas challenging mathematical items are computationally demanding, they continue to be solvable by applying known processes. More challenging verbal items—unscrambling long, unfamiliar words, for instance—may need trial and error, guessing, or memorized vocabulary. Respondents will be more likely to become stuck or slow down on such items, producing greater latency than with more mechanical arithmetic reasoning.

Based on the results of the sequential tests, we remove the non-significant two-way interactions $\alpha\gamma$ and $\alpha\delta$, and retain only the significant interaction $\gamma\delta$. We denote this model as Model 3. Given the result in Figure 6, we found that the Participant block effect is not significant, so we remove this block and obtain our final model:

$$Y_{ijklmn} = \mu + \alpha_i + \beta_{j(i)} + \gamma_k + \delta_l + \tau_m + (\gamma\delta)_{kl} + \varepsilon_{ijklmn}$$

The final ANOVA results confirm the significance of several key effects in explaining variation in task completion time. All main effects are highly significant ($p < 0.0001$), indicating that each factor independently contributes to task performance. We take the Time Interval block as an example for the significance testing at this stage for its random nature:

$$\begin{cases} H_0 : \sigma_\tau^2 = 0 \\ H_A : \sigma_\tau^2 > 0 \end{cases}$$

Based on Figure 3, the F-statistic for Time Interval is 18.93 with a p-value less than 0.0001. This provides strong evidence against the null hypothesis. Therefore, the random effect of Time Interval is statistically significant.

	Source	DF	Type III SS	Mean Square	F Value	Pr > F
*	MusicType	1	2696.472006	2696.472006	727.88	<.0001
	Volume(MusicType)	4	1827.405819	456.851455	123.32	<.0001
*	TaskType	1	1243.259113	1243.259113	335.60	<.0001
*	Difficulty	1	3088.980000	3088.980000	833.84	<.0001
	TimeInterval	2	140.248252	70.124126	18.93	<.0001
	TaskType*Difficulty	1	185.730689	185.730689	50.14	<.0001
	Error: MS(Error)	277	1026.158609	3.704544		
* This test assumes one or more other fixed effects are zero.						

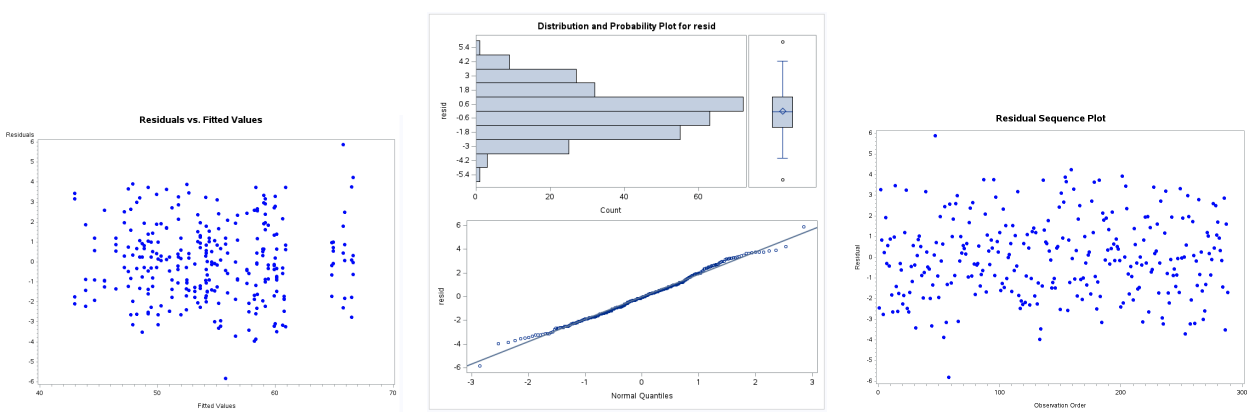
Figure 3: Final Model ANOVA Table

3.3 Diagnostics

The plot of residuals vs. fitted values is symmetric and evenly spread over the interval of fitted values, with no obvious funnel shape or curvature. This confirms that the constant variance assumption is adequately met, and there is no strong graphical evidence of heteroscedasticity or model misspecification.

All of the Shapiro-Wilk, Kolmogorov-Smirnov, Cramer-von Mises, and Anderson-Darling tests produce a p-value of greater than 0.05, which is an indication that residuals do not significantly stray from normality. The Q-Q plot is consistent with this finding, and all of the residuals appear nearly on the theoretical normal line, with only minimal deviation toward the extremes.

The residual sequence plot demonstrates neither a systematic trend nor autocorrelation, with residuals randomly spread out over observation order. This is an indication that the independence assumption is well satisfied and that no temporal patterns or order effects appear among residuals.



(a) Constant Variance Check (b) Normality Check (c) Independence Check

Figure 4: Diagnostic Plots for the Final Model

3.4 Remedial Measures

No remedial measures are needed, as all diagnostic checks—including normality, homoscedasticity, and independence—are adequately satisfied.

3.5 Task Completion Time Minimization

To minimize the task completion time, we compare all pairwise means of the six BGM treatment combinations while controlling the family-wise error rate using Tukey's procedure.

Let μ_i and μ_j denote the true mean task completion times under BGM treatment combinations i and j, where $i, j \in \{1, \ldots, 6\}$ index the six MusicType $\times$ Volume conditions. For example, $i = 1$ may correspond to the BGM condition `Instrumental+0%`, while $j = 6$ corresponds to `Lyrics+100%`. The family of hypotheses tested is:

$$\begin{cases} H_0 : \mu_i = \mu_j & \text{for all } i \neq j \\ H_A : \mu_i \neq \mu_j & \text{for some } i \neq j \end{cases}$$

Tukey-adjusted comparisons reveal that the combination of Instrumental music at 50% volume results in the shortest task completion time (LSMean = 48.37), which is significantly lower than all other BGM settings. Specifically, it performs better than both silence, full-volume instrumental music, and all Lyrics-based conditions, with all pairwise p-values less than 0.0001. Additionally, the boxplot on the right further confirms this result.

For a more detailed view of the confidence intervals, see Figure 8.

Least Squares Means
Adjustment for Multiple Comparisons: Tukey

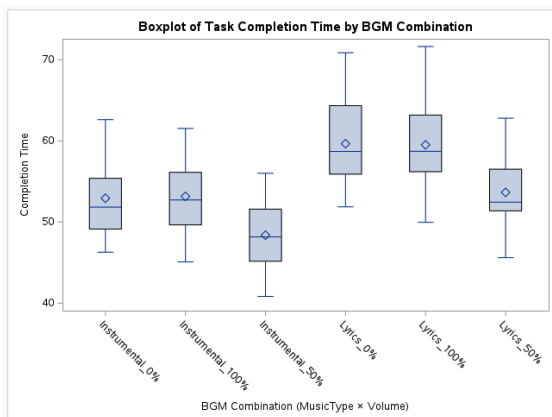
Volume	MusicType	Time LSMEAN	LSMEAN Number
0%	Instrumental	52.9316667	1
100%	Instrumental	53.1625000	2
50%	Instrumental	48.3743750	3
0%	Lyrics	59.6562500	4
100%	Lyrics	59.5185417	5
50%	Lyrics	53.6529167	6

Least Squares Means for effect Volume(MusicType)
Pr > |t| for H0: LSMean(i)=LSMean(j)

Dependent Variable: Time

i/j	1	2	3	4	5	6
1		0.9918	<.0001	<.0001	<.0001	0.4445
2	0.9918		<.0001	<.0001	<.0001	0.8125
3	<.0001	<.0001		<.0001	<.0001	<.0001
4	<.0001	<.0001	<.0001		0.9993	<.0001
5	<.0001	<.0001	<.0001	0.9993		<.0001
6	0.4445	0.8125	<.0001	<.0001	<.0001	

(a) Tukey's Method



(b) Boxplot of Time by BGM Combination

Figure 5: Tukey's Procedures

4 Conclusion

4.1 Conclusion and Final Comments

The study aims to analyze whether the characteristics of background music, specifically music type (instrumental vs. lyrical) and playback volume (0% vs. 50% vs. 100%), facilitate or hinder cognitive performance on tasks of different types and difficulties. Using a nested design (music volume nested under music types), we draw several conclusions:

Among all six combinations of music type and volume, instrumental music at 50% volume led to the shortest task completion times. This condition outperformed all others, including silence, full-volume instrumental music, and all lyrical music settings, with statistically significant differences (Tukey-adjusted p-values < 0.0001).

Each of the main factors (music type, volume, task type, and task difficulty) had a statistically significant impact on performance time (p-value < 0.0001). This suggests that cognitive performance is sensitive to background music and the nature of the task.

The task type $\times$ difficulty interaction shows a significant impact on performance time (p-value < 0.0001), indicating that the effect of difficulty is not uniform between task types. In particular, the increased difficulty led to a greater performance degradation for verbal tasks than for math tasks.

Although participants were treated as a fixed block factor, their effect on performance time is statistically insignificant (p-value $= 0.5725$) and subsequently removed from the final model. The time interval is modeled as a random block effect, and its significant effect addresses the timing of the exercise when measuring cognitive performance (p-value < 0.0001).

4.2 Potential Problems

Although the group carefully designed the experiment, there are several limitations:

1. **Measurement Error in Time:** Due to the manual adjustments of multiple experimental factors between trials (music type, volume), slight timing inaccuracies may have occurred when recording task completion times. The frequent switching of music types and volume levels within a short time period may produce cognitive effects that differ from more naturalistic settings, where music tends to be more consistent and stable.

2. **Lack of Accuracy Assessment:** Our group asked participants to make a consistent effort and try their best to complete the tasks. We did not evaluate whether participants answered the tasks correctly.

3. **Participant Fatigue:** Each participant completed a large number of tasks per session (48 trials per time interval). As the session progressed, fatigue or reduced concentration may have affected performance, potentially confounding the effects of background music.

4. **Limited Sample Size:** The experiment involved only two participants, each required to complete a large number of tasks. This not only limits the size of the dataset but may also introduce fatigue-related bias.

4.3 Suggested Improvements

To enhance the results of the experiment, we recommend the use of software that synchronizes task presentation with background music playback. This would eliminate delays caused by manual intervention and ensure that task timing begins only after the correct audio settings are fully applied.

Furthermore, the experiment should be divided into multiple shorter sessions conducted across different days. This would help reduce participant fatigue and maintain consistent performance levels. Finally, increasing the number of participants would strengthen the robustness of statistical inference, allow for more accurate estimation of individual variability.

5 Appendix

5.1 Omitted Figures

	Source	DF	Type III SS	Mean Square	F Value	Pr > F
*	MusicType	1	2696.472006	2696.472006	616.94	<.0001
	Volume(MusicType)	4	1827.405819	456.851455	104.52	<.0001
	TaskType	1	1243.259113	1243.259113	284.45	<.0001
	Difficulty	1	3088.980000	3088.980000	706.74	<.0001
	Participant	1	1.193512	1.193512	0.27	0.6017
	TimeInterval	2	140.248252	70.124126	16.04	<.0001
	Error: MS(Error)	277	1210.695785	4.370743		

* This test assumes one or more other fixed effects are zero.

Figure 6: Model 3 ANOVA Table

Tests for Normality				
Test	Statistic		p Value	
Shapiro-Wilk	W	0.993197	Pr < W	0.2163
Kolmogorov-Smirnov	D	0.034015	Pr > D	>0.1500
Cramer-von Mises	W-Sq	0.06393	Pr > W-Sq	>0.2500
Anderson-Darling	A-Sq	0.515332	Pr > A-Sq	0.1987

Figure 7: Normality Tests

Least Squares Means for Effect Volume(MusicType)				
i	j	Difference Between Means	Simultaneous 95% Confidence Limits for LSMean(i)-LSMean(j)	
1	2	-0.230833	-1.358318	0.896651
1	3	4.557292	3.429807	5.684776
1	4	-6.724583	-7.852068	-5.597099
1	5	-6.586875	-7.714359	-5.459391
1	6	-0.721250	-1.848734	0.406234
2	3	4.788125	3.660641	5.915609
2	4	-6.493750	-7.621234	-5.366266
2	5	-6.356042	-7.483526	-5.228557
2	6	-0.490417	-1.617901	0.637068
3	4	-11.281875	-12.409359	-10.154391
3	5	-11.144167	-12.271651	-10.016682
3	6	-5.278542	-6.406026	-4.151057
4	5	0.137708	-0.989776	1.265193
4	6	6.003333	4.875849	7.130818
5	6	5.865625	4.738141	6.993109

Figure 8: Tukey's Method: Confidence Intervals

5.2 SAS Code

```
/* Importing Data */
proc import datafile="/home/u64156445/Final Project/BGM_data_final.xlsx"
    out=bgm_data dbms=xlsx replace;
    sheet="Sheet1";
    getnames=yes;
run;

/* Model 1 Initial Model */
proc glm data=bgm_data;
    class MusicType Volume TaskType Difficulty Participant TimeInterval;
    model Time = MusicType Volume(MusicType) TaskType Difficulty
       Participant TimeInterval
                 MusicType*TaskType MusicType*Difficulty
                    TaskType*Difficulty
                 MusicType*TaskType*Difficulty / clparm;
    random TimeInterval / test;
run;

/* Model 2 */
proc glm data=bgm_data;
    class MusicType Volume TaskType Difficulty Participant TimeInterval;
    model Time = MusicType Volume(MusicType) TaskType Difficulty
       Participant TimeInterval
                 MusicType*TaskType MusicType*Difficulty
                    TaskType*Difficulty / clparm;
    random TimeInterval / test;
run;

/* Model 3 */
proc glm data=bgm_data;
    class MusicType Volume TaskType Difficulty Participant TimeInterval;
    model Time = MusicType Volume(MusicType) TaskType Difficulty
       Participant TimeInterval / clparm;
    random TimeInterval / test;
run;
```

```
/* Model 4 (final model) */
proc glm data=bgm_data;
    class MusicType Volume TaskType Difficulty Participant TimeInterval;
    model Time = MusicType Volume(MusicType) TaskType Difficulty
       TimeInterval
                 TaskType*Difficulty / clparm;
    random TimeInterval / test;
run;

/* Diagnostics */
/* 1. Constant Variance Check */
proc glm data=bgm_data;
    class MusicType Volume TaskType Difficulty Participant TimeInterval;
    model Time = MusicType Volume(MusicType) TaskType Difficulty
       TimeInterval
                 TaskType*Difficulty;
    random TimeInterval / test;
    output out=resids r=resid p=yhat;
run;

goptions reset=all;
symbol1 v=dot c=blue h=1.2;
axis1 label=('Residuals');
axis2 label=('Fitted Values');

proc gplot data=resids;
    plot resid*yhat / vaxis=axis1 haxis=axis2;
    title 'Residuals vs. Fitted Values';
run;

/* 2. Normality Check */
proc univariate normal plot data=resids;
    var resid;
    title 'Distribution of the Residuals';
run;

/* 3. Independence Check */
data resids2;
    set resids;
    order = _n_;
run;

symbol1 v=dot c=blue h=1;
axis1 label=(a=90 'Residual');
axis2 label=('Observation Order');

proc gplot data=resids2;
    plot resid*order / vaxis=axis1 haxis=axis2;
    title 'Residual Sequence Plot';
run;

/* Task Completion Time Minimization */
proc glm data=bgm_data;
```

```
    class MusicType Volume TaskType Difficulty Participant TimeInterval;
    model Time = MusicType Volume(MusicType) TaskType Difficulty
       TimeInterval
                 TaskType*Difficulty;
    random TimeInterval / test;
    lsmeans MusicType / pdiff cl adjust=tukey;
    lsmeans Volume(MusicType) / pdiff cl adjust=tukey;
    title 'Tukey-adjusted Comparisons for BGM Factors';
run;

/* Boxplot */
data bgm_data_combo;
    set bgm_data;
    BGM_Combo = catx('_', MusicType, Volume);
run;

proc sgplot data=bgm_data_combo;
    vbox Time / category=BGM_Combo;
    title 'Boxplot of Task Completion Time by BGM Combination';
    xaxis label='BGM Combination (MusicType * Volume)';
    yaxis label='Completion Time';
run;
```